

Data Science with Python Internship – Task 2

Task Description

In Task-1, you worked on the **Student Performance Dataset** and learned how to clean and analyze data.

Now, you will take one of the most famous datasets in Data Science – the **Titanic Dataset**. This dataset contains passenger details (age, class, gender, etc.) and whether they survived. Your task is to analyze survival patterns and visualize them.

What You'll Build (Requirements)

1. Load Titanic dataset from Kaggle.
2. Clean data (handle missing values like Age).
3. Answer questions such as:
 - Who survived more: males or females?
 - Did passenger class affect survival chances?
 - What was the survival rate by age group?
4. Visualize using **Seaborn / Matplotlib**:
 - Bar chart of survival by gender.
 - Bar chart of survival by class.
 - Histogram of passenger ages.

Why This Task Is Important

- Teaches you how to handle **real-world messy data**.
- Introduces **data visualization**, a key skill for Data Scientists.
- Helps you practice asking and answering **business questions with data**.

Deliverable

A Jupyter Notebook with:

- Data cleaning steps.
- Analysis answers.
- Graphs/plots with insights.

Sample Starter Code

```
python

import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("titanic.csv")

# Preview data
print(df.head())

# Survival rate by gender
sns.barplot(x="Sex", y="Survived", data=df)
plt.show()

# Survival rate by passenger class
sns.barplot(x="Pclass", y="Survived", data=df)
plt.show()

# Histogram of ages
df["Age"].plot(kind="hist", bins=20, title="Passenger Ages")
plt.show()
```

Helpful Resources

- Titanic Dataset – Kaggle
- [Pandas Documentation](#)
- [Seaborn Documentation](#)
- [Data Science Crash Course – FreeCodeCamp](#)

🔧 Tools

- Google Colab (free cloud Jupyter Notebook)
- Kaggle
- Python Libraries: Pandas, Matplotlib, Seaborn